



Advancing open science through research assessment reform? Content analysis of CoARA action plans

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Abstract

Research assessment reform and open science are frequently presented as aligned reform movements within science, with assessment reform expected to incentivize open science practices. However, empirical evidence of this integration remains limited. We conducted a quantitative content analysis of open science terminology in action plans from Coalition for Advancing Research Assessment (CoARA) signatories—institutions publicly committed to reforming research assessment. Analysing 248 publicly available action plans, we coded mentions of open science terms across values, umbrella, and operational categories. Results reveal skewed distributions of open science terms: most terms appear in fewer than 20% of documents. Values terms substantially outnumber operational terms in top mentions, suggesting organizations currently emphasize aspirational principles over concrete practices. Open Access emerges as the by far the most widely mentioned operational practice, while key activities emphasized in policy frameworks (Open Evaluation, Open Methods, Open Software) receive minimal attention. Overall, our findings suggest open science integration into assessment reform remains in early developmental stages, characterized by values-heavy discourse, and gaps between policy aspirations and organizational commitments. Beyond substantive findings, this study demonstrates how terminological analysis can track science reform movement development and provide data for longitudinal monitoring of open science and its convergence with reforms of research assessment.

Keywords Open science · Research assessment reform · Content analysis · CoARA

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Introduction

A transnational research assessment reform movement has emerged across academia in recent years, framing irresponsible evaluation practices as a shared problem requiring collective action. Arguably the Coalition for Advancing Research Assessment (CoARA) has become the largest and best known trans-national backbone initiative for coordinating such reforms. Supported by the European Commission, Science Europe, and the European Universities Association, this initiative has gathered (at the time of writing) over 900 signatories from organizations voluntarily agreeing to enact commitments and principles set out in its text, the Agreement on Reforming Research Assessment (CoARA, 2022).

CoARA has built on and expanded earlier influential statements like the San Francisco Declaration on Research Assessment (DORA, 2013), the Leiden Manifesto (Hicks et al., 2015) and the Metric Tide (Wilsdon, 2016), which raised concerns about inappropriate uses of quantitative indicators in research assessments and their adverse influences on the science system.¹ The scope of CoARA has though extended beyond a focused discussion on unintended consequences and technical limitations of specific performance indicators (Hicks et al., 2015), toward broader questions about what forms of knowledge, careers, and research cultures evaluations should reward (Curry et al. 2020, 2022; Wilsdon, 2021). This ‘responsible research assessment’ agenda has widened via the focus on responsible uses of evaluative bibliometrics converging with agendas of parallel science reform movements (Rushforth & Hammarfelt, 2023).

Notably, the current research assessment reform movement articulated by CoARA and similar initiatives has developed alongside the open science movement. Open science was initially associated with calls for open access publishing and open sharing of research data, but has subsequently expanded to encompass a much broader range of values and activities, including open evaluation, open research methods, citizen science, equity, diversity and inclusion (EDI or DEI) and more (UNESCO, 2021; Chtena et al., 2025; Ni & Waltman, 2024; Science Europe, 2024).

Multiple voices in both movements have cited alignments between research assessment reform and open science, with an oft mentioned strategic priority for assessment reform being to incentivize and thus help mainstream open science practices that current assessment systems do not tend to reward or recognize (European Commission et al., 2017; Friesike & Schildhauer, 2014; Magee, 2022). Specifically, assessment practices focused narrowly on scientific publications in ‘high impact’ journals have been identified as one of the key obstacles for advancement of open science practices (Amsterdam Call, 2016; Plan S, n.d.).

CoARA’s *Agreement on Reforming Research Assessment* (ARRA) (CoARA, 2022) itself incorporates many aspects of the open science agenda:

- Recognition of open science practices: Recognition of the broad diversity of... practices that contribute to robustness, openness, transparency, and the inclusiveness of research and the research process (Core commitment 1 scope)

¹ These were preceded by debates and interventions by bibliometricians through conferences, consultations, and debates in the pages of journals like *Scientometrics*. We acknowledge that CoARA has also drawn some criticisms from some bibliometricians (Abramo 2024) – but this debate is outside our scope.

- Transparency: Organisations should be clear and transparent about assessment processes and the tools and criteria they use, and make guidance on their assessment approaches openly available (Commitment 7 scope)
- Open research information: Ensure independence and transparency of the data, infrastructure and criteria necessary for research assessment and for determining research impacts (Principle for overarching conditions)
- Sharing data: Evaluate practices, criteria and tools based on solid evidence and the state-of-the-art in research on research, and make data openly available for evidence gathering and research (Commitment 10)

It is therefore to be expected that CoARA signatories might include some elements of open science in their reform agenda. But are organizations that are reforming their assessment practices engaging with open science? And if so, which aspects, and to what extent?

Despite rhetorical alignments between these two movements, empirical evidence about how organizations that have committed to assessment reform are incorporating open science remains limited. This study aims to reduce this gap by exploring what kinds of open science terms are mentioned within commitments that CoARA signatory organizations – as early responders to this leading reform initiative—have made publicly. To do so, we use content analysis to examine the open science language that signatory organizations express in their CoARA action plans. Action plans are a particularly useful resource, as the key formal media through which signatories set out their plans for local reforms in a publicly available forum – and have already become the focus of exploratory research (Malgarani et al., 2025; Nykyri & Niemi, 2024) and monitoring tools (CNRS, n.d.). According to CoARA’s website:

“In the context of the CoARA Agreement on Reforming Research Assessment, signatory organisations (this incl. also member organisations) are invited to share with the community how their organisation has started the process of implementing the Core Commitments according to an Action Plan with defined milestones within one year of signing the Agreement.” (CoARA, n.d.)

Our content analysis provides a timely means of mapping the commitments CoARA signatories are making to open science and gives provisional strategic insights into the current ‘readiness’ of signatory organizations to recognize and reward open science through assessment reforms.

Drawing on descriptive statistics, we seek to answer the following research questions:

1. How do CoARA signatories articulate open science commitments in their action plans?
2. Are values-based terms and umbrella terms more frequently and consistently emphasized than operational terms across COARA action plans?
3. Which open science terms are most widely occurring, and which are rarely or never mentioned?

As our analysis reveals, organizations are mostly emphasizing values language over operational language, suggesting open science integration into assessment reform remains still in early developmental stages.

Methods

CoARA action plans were downloaded from the CoARA Zenodo repository (<https://edu.nl/79pmv>), where they are publicly available, on December 12, 2025. When multiple versions from the same organization were available, we retained only the most recent version (for example, for Tampere University² and the University of Barcelona³). We excluded documents that were non-machine readable ($N=2$)⁴ or not in English ($N=2$).⁵ In addition, the University of Barcelona's monitoring report⁶ was excluded from the analysis, leaving a final total of 248 action plans, representing over a quarter of total CoARA signatories ($N \approx 912$) at the time of download.

A content analysis was performed using a keyword glossary (described in the next section and listed in full in Table 3) on the document set using R. Our text extraction procedure involved cleaning the documents to remove boilerplate text – namely the CoARA commitments, as 104 organizations had copy-pasted these into their action plans. This text was removed because it featured several open science key terms from our glossary, which we judged would give an inaccurate impression of open science engagement in these organizations.

Term extraction was performed using case-insensitive exact term matching. To avoid double-counting, the script was configured to ensure terms containing other terms (e.g. *Open Research Information* containing *Open Research*) were searched for as distinct entities. Where relevant, we included common variations in the glossary, such as hyphenated and non-hyphenated forms (e.g. ‘pre-print’ and ‘preprint’) and related word forms (e.g. ‘transdisciplinary’ and ‘transdisciplinarity’).

We generated both a binary dataset (1 – term appears at least once in a document; 0 – term does not appear in a document), and a frequency dataset (counting how many times each given term appears in each document). We also imported the following metadata: type of organization according to CoARA's classification of signatories and the country in which the organization is based according to CoARA (from CoARA Dashboard, n.d., an open resource developed by two of the authors). Finally we extracted word count data per document (excluding boilerplate). As the wordcounts showed skewed distribution, we decided to normalize frequency data to account for document length differences, using the formula:

$$\text{Normalized frequency} = (\text{term mentions} / \text{word count}) \times 1000$$

We calculated descriptive statistics for individual terms and thematic cluster groups to explore patterns in open science term prevalence across action plans.

² <https://zenodo.org/records/11399434>.

³ <https://zenodo.org/records/16616784>.

⁴ The action plans of Aalto University and the University of South-Eastern Norway.

⁵ <https://doi.org/10.5281/zenodo.17186429> and <https://doi.org/10.5281/zenodo.17192573>.

⁶ <https://zenodo.org/records/16267521>.

Coding framework

Constructing a robust framework of terms is crucial for content analysis as a deductive social science method. Clearly wide-ranging topics like open science can cause some controversy over the terms to include or exclude, meaning framework validity must always lie in defensibility rather than perfection (Rourke & Anderson, 2004). We took several established steps to build our glossary of open science terms: reviewing search terms used in previous studies (e.g. by Marangoni et al., 2025; Ross-Hellauer et al., 2022) and terms featured in authoritative statements (e.g. UNESCO, 2021; Science Europe, 2024), drawing on our own knowledge as participants in these reform communities, and subjecting the glossary to external validation by two open science policy specialists in our respective countries (although responsibility for any shortcomings are our own).

This iterative process produced a list of individual terms, which we organized into two grouping systems: 1) thematic clusters and 2) values-operational-umbrella categories (see Table 3).

Thematic clusters: Based on our reading of the literature and field experience, we grouped related terms within the open science landscape. For example, the individual terms *Open Access*, *Open Scholarly Communication*, and *Preprints* were grouped under *Open Publishing*. This clustering allows us to compare areas of open science activity and guards against hasty conclusions from individual keywords. For instance, while *Team Science* occurs infrequently as a standalone term, related terms like *Collaboration* appear more often, preventing us from hastily concluding that organizations are completely uninterested in Team Science.

Values-operational-umbrella categories: Inspired by the values mapping literature (Bozeman, 2007), we distinguish between values-based terms and the broad range of operational (practice-based) terms from *Data Management* to *Open Educational Resources* to *Community Engagement*. We also identified a third category, umbrella terms.

Values terms focus on high-level principles around which there is considerable societal consensus (Bozeman, 2007). These operate at a level of high generality, often captured by single words like equity, transparency, or inclusion. Open science is deeply embedded in values discourse (Deutsche Forschungsgemeinschaft 2022; UNESCO, 2021; Science Europe, 2024), and many of our values terms were drawn directly from authoritative statements by UNESCO and Science Europe.

Operational terms refer to diverse practices, methods, infrastructures, policies, tools, and indicators for realizing open science objectives. Put simply, if values terms represent the ends of open science, operational terms represent the means. In many cases, a single value corresponds to multiple operational approaches (Bozeman, 2007). For example, transparency might be pursued through preregistration or open peer review. Conversely, some operational terms may serve multiple values simultaneously.

Umbrella terms group together disparate concerns into a recognizable social movement, providing a common focus and traveling buzzwords that addresses multiple audiences, supporting movement branding and institutional agenda-building (Rip & Voß, 2019). *Open Science* and *Open Research* are examples.

Introducing this higher-order categorizations helps us to organize an otherwise diffuse keyword set into a framework that supports theoretically-informed exploration of patterns across the action plans. The three-part framework enables us to examine:

- Whether organizations emphasize broad value commitments or specific operational mechanisms.
- Whether umbrella terms are mentioned more frequently than specific values or operational approaches (and which umbrella terms are more popular).
- How different thematic clusters (e.g. *Open Publishing* vs. *Societal Engagement*) vary in prevalence.

Results

The analysis draws on 248 action plans, submitted by CoARA signatories from across 35 countries and 19 trans-national organizations (which CoARA classify as being either ‘European’ or ‘Global’). After grouping countries by region, we found submissions originated relatively evenly from Nordic (23%), Southern European (24%), and Western/Central European (21%) countries, with relatively fewer in total from Eastern European countries (15%). Among signatories, universities constituted 66% ($N=164$), research centres and funders 10.5% each ($N=26$), academies and learned societies 6.5% ($N=16$), research assessment agencies 3.6% ($N=9$), and other non-profits 2.8% ($N=7$) (Table 6B). To examine whether organizational mission shapes terminological choices, we conducted a cross-tabulation analysis comparing term mentions across organization-type (this data can be found in the supplementary materials). This analysis revealed no strong discernible patterns distinguishing universities from funders, learned societies, and other organizational categories. Most terms showed similar mention rates across organization types, including those where mission-oriented differences might be hypothesized (e.g. *Open Access* being concentrated more among funders due to policy mandates). In light of these findings and the university-dominated sample, our analysis subsequently focuses on patterns of terms aggregated across all signatory types.

Across the action plans, open science terminology displays a highly skewed distribution. Each term appears in an average of 24 documents (median=6 documents), and only 10 of the 73 terms are present in more than 20% of plans (Table 1). Figure 1 illustrates this skewness.

Of the umbrella terms, *Open Science* is the most commonly mentioned across action plans. The term appears in 167 documents (68%), compared to the alternative umbrella terms *Open Research* (20%) and *Open Scholarship* (2%) (Table 4).

Breadth and depth rankings reveal consistent patterns: the most widely used terms are predominantly values-based. Binary analysis shows that of the top ten terms by document

Table 1 Descriptive statistics for open science term distribution across CoARA action plans – binary data

Statistic	Value
Number of documents	248
Number of open science terms	73
Mean documents per term	24
Median documents per term	6
Standard deviation	38
Range	0–167
Terms in > 20% of documents	10

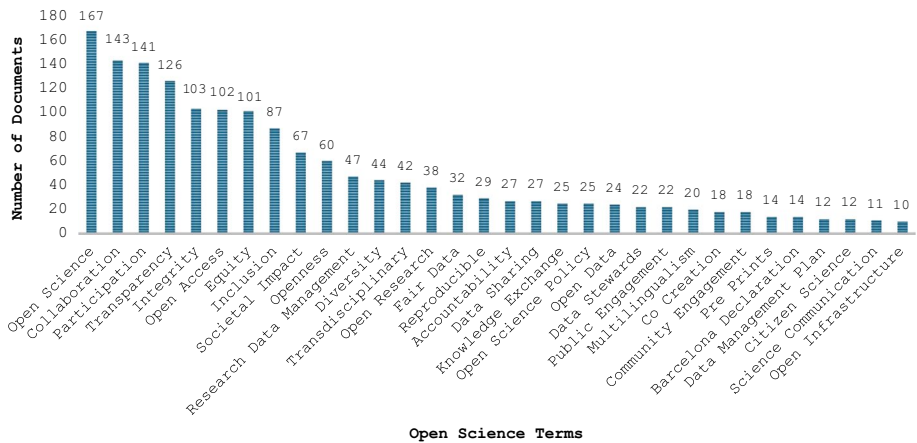


Fig. 1 Open science term mentions in CoARA Action Plans. Using binary data, shows number of documents (out of Total $N=248$) in which each term appears at least once. Only terms that appear in ≥ 10 documents are displayed

coverage, eight are values terms, one is an operational term (*Open Access*), and one an umbrella term (*Open Science*) (Table 4). Frequency analysis confirms this pattern. When mentions are normalized by document word count and summed up across all documents, values terms (e.g. *Collaboration*, *Participation*, *Transparency*, *Integrity*) dominate the upper ranks (Table 5). *Open Science* has the highest aggregated normalized frequency (459), more than double that of the next-ranked term, *Collaboration* (185). Sixty-seven percent of glossary terms have aggregated normalized frequencies below 10 across the entire dataset (Table 5). 13 operational terms showed zero mentions across all 248 action plans, including *Open Notebooks*, *Protocol Sharing*, and *Crowdsourcing* (Table 4).

Grouping terms into thematic clusters further clarifies the stratification between values and operational language (see Table 3 for which terms are assigned to a cluster). As shown in Table 2, the top quartile is composed exclusively of values-based and umbrella clusters: *Collaboration* (80%), *Umbrella Terms* (71%), *Research Integrity Values* (65%), and *Equity, Diversity and Inclusivity* (60%). No operational clusters appear in this top tier. The operational clusters *Open Publishing* (43%), and *Societal Engagement Activities* (41%), and *Open Data* (39%) appear most widely, but none are present in a majority of action plans. Lower-ranked clusters consist almost exclusively of operational topics, such as *Open Source Software*, *Open Evaluation*, and *Open Education*.

Examining individual operational terms reveals further differentiation. *Open Access* appears most broadly across documents (102 plans), followed by *Research Data Management* (47 documents, 19%) and *FAIR Data* (32 documents, 13%). Mentions of specific operational terms like *Open Peer Review* (7 documents, 3%), *Open Source Software* (3 documents, ~1%), and *Open Methods* (1 document, ~0%) are very sparse (Table 4).

Several terms prioritized in major international frameworks, such as UNESCO's Recommendations on Open Science, generally appear infrequently in CoARA action plans. *Citizen Science* (individual term) appears in only 5% of documents (Table 4), and the *Open Source Software*, *Open Education*, and *Open Evaluation* thematic clusters in 4% of plans (Table 2). These patterns indicate that while certain focuses (particularly **Open Access**)

Table 2 Open Science thematic cluster ranked by occurrences across CoARA action plans

Quartile	Rank	Thematic cluster	Document mentions ($N=248$)	Document mentions (%)	Category
1st (top)	1	Collaboration	199	80%	Value
	2	Umbrella Terms	175	71%	Umbrella
	3	Research Integrity Values	160	65%	Value
	4	Equity Diversity and Inclusivity	150	60%	Value
2nd	5	Open Publishing	106	43%	Operational
	6	Societal Engagement Activities	101	41%	Operational
	7	Open Data	97	39%	Operational
	8	Societal Impact	67	27%	Value
3rd	9	Openness	60	24%	Value
	10	Research Integrity Interventions	33	13%	Operational
	= 11	Open Research Information	25	10%	Operational
	= 11	Open Science Policy	25	10%	Operational
4th (bottom)	13	Multi lingualism	20	8%	Operational
	14	Open Science Indicators	16	6%	Operational
	15	Open Education	11	4%	Operational
	16	Open Source Software	10	4%	Operational
	17	Open Evaluation	9	4%	Operational

Using binary data, shows number of documents in which each thematic cluster appears at least once. Umbrella category refers to terms like open science, open research, and open scholarship

receive attention, many operational dimensions of open science are currently marginal in institutional action plans.

Discussion

The skewed distribution of open science terminology across the action plans is reflected in the uneven appearances of terms across documents and the small subset of terms appearing in more than 20% of documents. This suggests that many open science dimensions remain in an emergent and exploratory stage among CoARA signatories.

The relative immaturity of open science's embedding in action plans helps shed light on another key finding: the higher prevalence of values and umbrella terms compared to operational-based open science terms (see rankings in Tables 4, 5). The thematic cluster analysis reinforces and extends these individual term findings, further revealing stratification between values and operational language. The complete absence of operational clusters from the top quartile, combined with their exclusive presence in the bottom quartile, underscores this interpretation. Furthermore, even the most successful operational clusters (*Open Publishing*, *Societal Engagement*, *Open Data*) achieved only moderate adoption (39–43% of documents), while values clusters routinely appeared in a majority of action plans (Table 2). We infer several potential explanations of this pattern. First, when a field lacks established operational practices, organizations naturally emphasize abstract values over concrete operational mechanisms. Second, values language generally enables broad

coalition-building without requiring broad consensus on specific practices, with achievement of the latter being a major challenge in organizational reforms. Third, values commitments require fewer organizational resources than operational commitments and given their widespread circulation in research governance already, are easier to commit to and may already be embedded in institutional discourse. Equally, this emphasis on values may be appropriate for the developmental stages of research assessment reform that action plans tend to capture, as establishing shared values should be a necessary prerequisite for negotiating operational details of evaluation (re)design (Himanen et al., 2024). Encouragingly for those seeking to promote open science through assessment reform, the majority of open science terms—including most operational terms—are mentioned at least somewhere across the dataset, which may be a reasonable foundation for a still nascent reform movement. The outstanding question is therefore whether current aspirational language represents an early developmental stage that will eventually yield implementation, or whether it substitutes for more difficult operational changes that may not materialize?

Open Access stands out as an operational term with broad coverage, reflecting its relative maturity within the open science movement (tallying with recent survey evidence by Hyrkkänen et al., 2023). While this demonstrates attention to at least one dimension of open science, it raises a strategic concern: the risk that open science becomes synonymous with open access alone, obscuring other critical dimensions. This concern is amplified by our finding that terms emphasized in UNESCO's comprehensive framework—such as *Open Evaluation*, *Open Methods*, and *Open Software*—showed very low frequencies in action plans. This gap suggests a current disconnect between authoritative international policy frameworks and organizational language choices. We cannot say with certainty from content analysis data that organizations are de-prioritizing these more operational practices (although it is in our view likely they are), but it does at the very least suggest there is a terminological mismatch between policy discourse and institutional commitments.

Study strengths and limitations

Content analysis detects explicit mentions of pre-defined, fixed terms. From these mentions, we drew proxy indicators (aggregated binary and frequency data) to measure breadth and depth of engagement with various open science elements. Of course this method cannot capture context-dependent meanings in how terms are deployed, and while we can draw inferences from data on operational language, we cannot observe actual implementation in reforming organizations. As such, our exploratory approach may complement but cannot replace in-depth, longitudinal and partly qualitative research and evaluation on organizational implementation of assessment reforms and open science.

Although CoARA has arguably become the most developed and invested in research assessment reform initiative to date, CoARA is currently a predominantly European initiative, meaning dataset mostly reflects European institutional contexts (see Table 6A, for country-level comparisons of the organizations that have deposited action plans; and for a broader overview of CoARA signatories, see Pölönen & Rushforth, 2025). Though this European-concentration is a feature of the policy environment rather than a limitation of our sampling strategy, it does mean that findings should be interpreted largely as a read on current European reform dynamics rather than global patterns.

Validity questions in deductive versions of content analyses typically focus on framework adequacy (Rourke & Anderson, 2004). As with any theoretical construct, readers may question which terms should be included or excluded as such choices directly impact statistical results. While our framework is non-exhaustive, it was developed systematically from major policy documents and validated through external audit. The glossary focuses on standard terminology validated by open science experts. While some alternative phrasings of operational terms may exist, we are confident it captures the primary vocabulary used in institutional discourse on open science and that rare variations would not materially affect our analysis. Following content analysis conventions, we implore the coding framework to be judged on reasonableness rather than perfection (Rourke & Anderson, 2004).

Content analysis does offer some distinctive advantages. The method is unobtrusive and uses naturalistically occurring (and in this instance publicly available) data. We therefore avoided the research artifact phenomenon common in surveys and interviews, where organizational respondents may overestimate their open science actions to impress researchers or manage institutional reputation.

While our analysis covers only a sample of all CoARA signatories, it examines near enough the complete corpus of publicly available action plans, submitted at the time of analysis. Given that fewer than one-third of signatories have published action plans currently, these early submitters may be read as being at the more enthusiastic and committed end of the signatory spectrum. If the presumably motivated early adopters show somewhat limited and uneven engagement with open science terminology so far, later or less enthusiastic adopters may be less advanced still. Our findings therefore are unlikely to be underestimating current engagement levels across all the CoARA signatories – lending further credence to the robustness of our conclusions.

Conclusion

Our content analysis of CoARA action plans suggests a still uneven and overall developing patterns of engagement with open science among organizations setting out on their research assessment reform journeys. The categorical distinction between values, umbrella terms, and operational dimensions suggests that open science may currently align more easily with research assessment reform plans at the values and umbrella-term level. The values that responsible research assessment aspires to realize appear consistent with values associated with open science broadly conceived (e.g. UNESCO Recommendations and Science Europe's articulations of open science values). However, notwithstanding aforementioned limitations in manifest content analysis, overall more operationally-oriented language appears less frequently, with important exceptions like **Open Access**. Our findings reveal fragmented rather than absent engagement with operational language: most operational terms are mentioned somewhere, but no single operational term occurs in a majority of documents. That both binary analyses and frequency analyses support this pattern reinforces the finding.

These findings underscore the value of terminological analysis for understanding reform movement development. Language choices reveal priorities, signal commitments, and may

provide insights into what transformations currently seem possible to organizational signatories. By mapping open science terminology systematically, we can track convergences between open science and research assessment reform from aspirational statements toward operationalization – and over time begin to identify where aspirations may persistently substitute for more difficult implementation measures. This monitoring function becomes particularly important as the field matures and stakeholders assess whether initial commitments translate into substantive changes in practice (see the recent chorus of calls to evidence open science: GRIOS, 2025; Marangoni et al., 2025; OSML, 2025). We encourage future researchers to adapt and update our glossary as the field of open science evolves.

How might these results influence further development of these reforms? First, we hope to spark discussion about strategic priorities and debates regarding the relationship between open science and research assessment reform. This is timely, given lively debates *within* the reform community regarding the role that ought to be assigned to research assessment reforms in mainstreaming open science interventions may not be as settled as initially appeared. Alongside prominent calls to start rewarding open science to affect culture change (e.g. Nosek, 2019), recent interventions by prominent reform advocate Elizabeth Gadd questioned whether evaluating open science should be a strategic priority for research assessment, and if so, on what basis: as achievement markers to be rewarded and incentivized or as a bare minimum ‘hygiene factor’ for research(ers) to meet (Gadd, 2025)? Our finding can provide empirical inputs into these foundational normative inter- and intra-movement debates.

Beyond these substantive insights, methodologically our content analysis approach, openly available lexicon of terms, and conceptual framework provide a reproducible toolkit that can be adapted for longitudinal tracking of developments in how open science manifests in research assessment reform contexts, enabling analysts, supporters, sponsors, and other interested parties to monitor these developments as they unfold.

Appendix

See Tables 3, 4, 5, 6.

Table 3 Glossary of open science terms

Theoretical category	Cluster family	individual term	Variations
Umbrella terms	Umbrella	Open Science	Open-Science
		Open Research	Open-Research
		Open Scholarship	Open-Scholarship
Value terms	Equity, diversity and inclusivity	Open Knowledge	Open-Knowledge
		Equity	Equality
		Inclusion	Inclusivity; Under Represented; Under-Represented; Underrepresented
		Diversity	Diverse Contributions; Widening Contributions; Language Diversity; Epistemic Diversity; Indigenous Knowledge; Traditional Knowledge
		Transparency	Transparent Research
Research integrity	Collaboration	Integrity	–
		Accountability	–
		Participation	Participatory
		Collaboration	–
		Co-Creation	Cocreation; Co creation; Co-Created; Cocreated; Co-create; Co create; Co-Research; Coresearch; Co research
Operational terms	Openness	Team Science	–
		Openness	–
		Societal Impact	Societal Relevance
		Diversity Indicators	–
		Non Traditional Outputs	Non-Traditional Outputs; Nontraditional Outputs
	Multi lingualism	Altmetrics	Alt Metrics; Alt-Metrics; Alternative Metrics
		Open Science Indicators	–
		Multi Lingualism	Multilingualism; Multi-Lingualism; Multilingual; Multi lingual
		Helsinki Initiative	- Helsinki Initiative on Multilingualism in Scholarly Communication

Table 3 (continued)

Theoretical category	Cluster family	individual term	Variations
Open data		Data Sharing	–
		Open Data	Open-Data
		Fair Data	Fair Principles
		Care Principles	–
		Data Repositories	Data Repository
		Data Stewards	Data Steward; Data Stewardship
		Data Management Plan	DMP; Research Data Management Plan
		Research Data Management	Data Management; RDM
		Open Access	Open-Access
		Pre Prints	Preprints; Pre-Prints; Preprint; Pre-Print; Pre Print; Pre Printing; Preprinting; Pre-Printing
Operational terms	Research integrity interventions	Open Repository	Open-Repository; Open Repositories;
		Public Access	–
		Publishing Ethics	–
		Open Scholarly Communication	–
		Reproducible	Reproducibility
		Replication Studies	Replication Study
		Pre-Registration	Preregistration; Pre Registration; Pre-Registering; Pre-registering; Pre-Registered; Pre Registered; Preregistered;
		Protocol Sharing	–
		Open Methods	Open-Methods
		Open Notebooks	Open-Notebooks; Open Notebook
Open Evaluation		Open Science Badges	Open-Science Badges; Open Science Badge; Open-Science Badge
		Reporting Guidelines	–
		Open Evaluation	Open-Evaluation
		Open Peer Review	Open-Peer Review; Open Peer-Review

Table 3 (continued)

Theoretical category	Cluster family	individual term	Variations
Open source software		Open Source	Open-Source; Opensource
		Open Source Software	Open-Source Software; Opensource Software; OSS
		Open Software	Open-Software
		Open Code	Open-Code
		Open Hardware	Open-Hardware
		Open Tools	Open-Tools
Open Research Information		Open Research Information	Open-Research Information
		Barcelona Declaration	Barcelona Declaration on Open Research Information
		Open Infrastructure	Open-Infrastructure
		Open Dashboard	Open-Dashboard
		Open Educational Resources	Open Education Resources; OER
Open Education		Open Educational Practices	OEP
		Open Education	–
		MOOC	Massive Open Online Course; MOOCs

Table 3 (continued)

Theoretical category	Cluster family	individual term	Variations
Operational terms	Societal engagement activities	Science Communication	–
		Science Popularisation	Science Popularization
		Citizen Science	–
		Crowdsourcing	Crowd Sourcing; Crowd-Sourcing
		Public Engagement	–
		Community Engagement	–
		Stakeholder Engagement	–
		Knowledge Exchange	–
		Knowledge Brokering	–
		Transdisciplinary	Trans Disciplinary; Trans-Disciplinary; Transdisciplinarity; Trans Disciplinarity
		Open Science Policy	Open-Science Policy; Open Science Policies
		Open Science Officers	Open Science Officer; Open-Science Officer; Open-Science Officers
		Open Science Awareness	–

Table 4 Distribution of open science terminology ranked by document coverage in CoARA action plans – binary data

Rank	Term	Document appearances (<i>N</i> = 248)	Document appearances (%)
1	Open Science	167	67%
2	Collaboration	143	58%
3	Participation	141	57%
4	Transparency	126	51%
5	Integrity	103	42%
6	Open Access	102	41%
7	Equity	101	41%
8	Inclusion	87	35%
9	Societal Impact	67	27%
10	Openness	60	24%
11	Research Data Management	47	19%
12	Diversity	44	18%
13	Transdisciplinary	42	17%
14	Open Research	38	15%
15	Fair Data	32	13%
16	Reproducible	29	12%
= 17	Accountability	27	11%
= 17	Data Sharing	27	11%
= 19	Knowledge Exchange	25	10%
= 19	Open Science Policy	25	10%
21	Open Data	24	10%
= 22	Data Stewards	22	9%
= 22	Public Engagement	22	9%
24	Multilingualism	20	8%
= 25	Co Creation	18	7%
= 25	Community Engagement	18	7%
= 27	Pre Prints	14	6%
= 27	Barcelona Declaration	14	6%
= 29	Data Management Plan	12	5%
= 29	Citizen Science	12	5%
31	Science Communication	11	4%
32	Open Infrastructure	10	4%
33	Stakeholder Engagement	9	4%
= 34	Team Science	8	3%
= 34	Altmetrics	8	3%
36	Open Peer Review	7	3%
= 37	Open Research Information	6	2%
= 37	Non Traditional Outputs	6	2%
= 39	Open Repository	5	2%
= 39	Open Educational Resources	5	2%
= 39	Open Education	5	2%
= 39	Open Source	5	2%

Table 4 (continued)

Rank	Term	Document appearances (<i>N</i> = 248)	Document appearances (%)
= 43	Open Scholarship	4	2%
= 43	Data Repositories	4	2%
= 43	Pre Registration	4	2%
= 46	Helsinki Initiative	3	1%
= 46	Open Source Software	3	1%
= 48	Open Knowledge	2	1%
= 48	Open Science Indicators	2	1%
= 48	Open Scholarly Communication	2	1%
= 48	Open Evaluation	2	1%
= 48	Science Popularisation	2	1%
= 53	Replication Studies	1	0%
= 53	Open Methods	1	0%
= 53	Open Science Badges	1	0%
= 53	Open Software	1	0%
= 53	Open Code	1	0%
= 53	Open Hardware	1	0%
= 53	Mooc	1	0%
= 53	Knowledge Brokering	1	0%
= 61	Diversity Indicators	0	0%
= 61	Care Principles	0	0%
= 61	Public Access	0	0%
= 61	Publishing Ethics	0	0%
= 61	Protocol Sharing	0	0%
= 61	Open Notebooks	0	0%
= 61	Reporting Guidelines	0	0%
= 61	Open Tools	0	0%
= 61	Open Dashboard	0	0%
= 61	Open Educational Practices	0	0%
= 61	Crowdsourcing	0	0%
= 61	Open Science Officers	0	0%
= 61	Open Science Awareness	0	0%

Table 5 Terms ranked by frequency of mentions across all action plans – frequency data (normalized by document word length)

Rank	Term	Frequency
1	Open Science	459
2	Collaboration	185
3	Participation	171
4	Transparency	123
5	Open Access	112
=6	Integrity	101
=6	Equity	101
8	Inclusion	76
9	Societal Impact	62
10	Openness	49
11	Open Research	48
12	Research Data Management	45
13	Reproducible	31
14	Transdisciplinary	30
15	Diversity	25
16	Fair Data	21
= 17	Data Sharing	17
= 17	Knowledge Exchange	17
= 19	Open Data	15
= 19	Open Science Policy	15
= 21	Accountability	14
= 21	Multilingualism	14
= 21	Community Engagement	14
= 21	Public Engagement	14
25	Data Stewards	13
= 26	Co Creation	9
= 26	Barcelona Declaration	9
= 26	Data Management Plan	9
= 29	Citizen Science	8
= 29	Team Science	8
= 30	Altmetrics	7
= 30	Pre Prints	7
33	Stakeholder Engagement	6
= 34	Science Communication	4
4		
= 34	Open Educational Resources	4
= 34	Open Infrastructure	4
= 34	Open Peer Review	4
= 38	Open Source Software	3
= 38	Open Research Information	3
= 40	Non Traditional Outputs	2
= 40	Open Repository	2
= 40	Open Source	2
= 40	Open Education	2
= 40	Data Repositories	2
= 40	Pre Registration	2

Table 5 (continued)

Rank	Term	Frequency
= 40	Open Scholarship	2
= 40	Helsinki Initiative	2
= 48	Open Knowledge	1
= 48	Open Scholarly Communication	1
= 48	Science Popularisation	1
= 48	Open Science Badges	1
= 48	Open Evaluation	1
= 48	Open Hardware	1
= 48	Replication Studies	1
= 48	Open Software	1
= 48	Open Code	1
= 48	Open Science Indicators	1
= 58	Mooc	0
= 58	Knowledge Brokering	0
= 58	Open Methods	0
= 58	Diversity Indicators	0
= 58	Care Principles	0
= 58	Public Access	0
= 58	Publishing Ethics	0
= 58	Protocol Sharing	0
= 58	Open Notebooks	0
= 58	Reporting Guidelines	0
= 58	Open Tools	0
= 58	Open Dashboard	0
= 58	Open Educational Practices	0
= 58	Crowdsourcing	0
= 58	Open Science Officers	0
= 58	Open Science Awareness	0

Table 6 Regional and organization-type distributions of CoARA action plans

A. Regional distribution			
Region	Countries (Action plans per country)	Total (<i>N</i> = 248)	%
Southern Europe	Italy (28), Spain (28), Portugal (3), Andorra (1)	60	24.2
Nordic	Finland (29), Sweden (17), Norway (8), Denmark (4)	58	23.4
Western Europe	Germany (13), Netherlands (13), Switzerland (8), Austria (6), Belgium (6), France (5), Luxembourg (1)	52	21
Eastern Europe	Poland (9), Czechia (6), Hungary (6), Ukraine (4), Lithuania (4) Romania (2) Croatia (1), Slovenia (1), Estonia (1), Latvia (1), Moldova (1), Georgia (1), Bosnia and Herzegovina (1)	38	15.3
UK/Ireland	UK (12), Ireland (4)	16	6.5
Non-European	Australia (1), Brazil (1), Canada (1), Türkiye (1) Chile (1)	5	2
European Organizations	Europe (15)	15	6
Global Organizations	Global (4)	4	1.6
B. Organization type distribution*			
Organization type	Documents (n)	%	
Universities and their associations	164	66.1	
Research centres, infrastructures and their associations	26	10.5	
research funding organizations and their associations	26	10.5	
academies and learned societies	16	6.5	
Research assessment authorities/agencies	9	3.6	
other non-profit organizations	7	2.8	

*Groupings based on CoARA's classifications of signatories (<https://www.coara.org/agreement/signatories/>)

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Author contributions AR led project design, analysis, and wrote first draft NG provided data curation, supported figure preparation, writing and editing TS extracted data, writing and editing JP supported study design, writing and editing All authors approved the final manuscript.

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Data availability Data and code available at: DOI: [<https://doi.org/10.5281/zenodo.18268245>].

Declarations

Competing interests Janne Pölönen is the, forthcoming vice-chair of CoARA.

Ethical approval None required because the study is based on publicly available data.

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